SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:	03-08-2026	**Duration:**	60 minutes

Code of Conduct

I B. Nandini (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B. Nandini

Assignment Title:

Agriculture equipment rental and farm resource management platform

1. **Identify the Project Vision and Stakeholders**
    - Define the project's vision, objectives, and identify all key stakeholders involved.

Project Vision

To develop a digital platform that enables farmers to rent agricultural equipment, manage farm resources, and improve farming productivity by connecting equipment owners, farmers, and administrators through a simple and secure online system.

Project Objectives

- Provide online equipment rental services.
- Enable efficient farm resource management.
- Reduce equipment ownership costs for farmers.
- Increase equipment utilization for owners.
- Offer secure online booking and payment.
- Improve transparency through real-time tracking and notifications.

Key Stakeholders

Stakeholder	Role
Farmers	Rent equipment and manage farm resources
Equipment Owners	List equipment and receive rental income
Administrator	Manage users, equipment, bookings, and reports
Delivery Partner	Deliver and collect rented equipment
Payment Gateway	Process secure online payments
Agricultural Suppliers	Provide fertilizers, seeds, pesticides, etc.

2. **Create Epics and User Stories**
    - Break down the requirements into Epics and formulate well-structured User Stories using the format:
      As a <user>, I want <feature>, so that <benefit>.

Epic 1: User Management

User Story 1

As a farmer, I want to register an account so that I can rent farming equipment online.

User Story 2

As an equipment owner, I want to create my profile so that I can list my equipment for rent.

Epic 2: Equipment Management

User Story 3

As an equipment owner, I want to add equipment details so that farmers can view and rent them.

User Story 4

As a farmer, I want to search available equipment so that I can rent suitable machinery.

Epic 3: Booking Management

User Story 5

As a farmer, I want to book equipment for a selected date so that I can use it during my farming activities.

User Story 6

As an equipment owner, I want to approve or reject bookings so that I can manage my equipment schedule.

Epic 4: Payment Management

User Story 7

As a farmer, I want to pay online so that my booking is confirmed instantly.

Epic 5: Resource Management

User Story 8

As a farmer, I want to record fertilizers, seeds, labor, and expenses so that I can monitor farm resources.

Epic 6: Reports

User Story 9

As an administrator, I want to generate reports so that I can monitor platform performance.

3. **Define Acceptance Criteria**
    - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.

User Story 1

Given a new user visits the website

When valid registration details are entered

Then the account should be created successfully.

User Story 3

Given the equipment owner is logged in

When equipment details are submitted

Then the equipment should appear in the available equipment list.

User Story 5

Given equipment is available

When the farmer books the equipment

Then the booking should be confirmed and saved.

User Story 7

Given booking is confirmed

When payment is completed successfully

Then payment status should change to "Paid".

User Story 8

Given the farmer is logged in

When farm resources are entered

Then the system should save and display the updated records.

4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.

Priority	Product Backlog Item	Business Value
High	User Registration & Login	Essential
High	Equipment Listing	Essential
High	Equipment Search	Essential
High	Equipment Booking	Essential
High	Online Payment	Essential
Medium	Resource Management	High
Medium	Notifications	Medium
Medium	Booking History	Medium
Low	Ratings & Reviews	Low
Low	Reports & Analytics	Low

5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.

Sprint 1 User Stories

- User Registration
- User Login
- Equipment Listing
- Equipment Search
- Equipment Booking

Development Tasks

Registration Module

- Design registration page
- Validate user details
- Store user information

Login Module

- Create login page
- Authenticate users
- Manage sessions

Equipment Module

- Add equipment
- Edit/Delete equipment
- Display equipment list

Search Module

- Search by equipment type
- Search by location
- Filter by availability

Booking Module

- Book equipment
- Save booking
- Display booking status

6. **Estimate Effort Using Story Points**
    - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

User Story	Story Points	Justification
Registration	3	Simple forms and validation
Login	2	Authentication only
Equipment Listing	5	CRUD operations

User Story	Story Points	Justification
Equipment Search	5	Filtering and searching
Equipment Booking	8	Booking logic and validation
Payment Integration	8	API integration and security
Resource Management	5	Multiple data entries
Reports	3	Report generation

Estimation Technique Used: Fibonacci Sequence (2, 3, 5, 8)

7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Sprint Goal

Develop the core functionalities of the Agriculture Equipment Rental and Farm Resource Management Platform, enabling farmers to register, log in, search equipment, list equipment, and book available machinery.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	**Mark**
Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5
Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25